

CRYPTOGRAPHY FINAL PROJECT

Object

PKI REPORT

Date:

04/12/2025

Certificate Inspection

Sender signing certificate details:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Sender-Sign.cer -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 257 (0x101)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = PKI, CN = TEAM14ROOT, emailAddress = PKI.TEAM14@TEAM14.CA
        Validity
            Not Before: Dec  2 01:21:24 2025 GMT
            Not After : Mar  2 01:21:24 2026 GMT
        Subject: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = TEAM14, CN = TEAM14, emailAddress = SENDER@TEAM14.CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            Public-Key: (2048 bit)
            Modulus:
                00:b1:90:7b:ff:ce:77:c9:18:19:ed:a2:1c:aa:71:
                f6:b8:c7:7b:2e:68:9a:53:a0:52:d0:74:aa:00:7c:
                b6:95:72:87:be:58:8b:62:ef:8e:13:59:23:8f:24:
                a7:a2:79:cd:f7:cb:c3:eb:66:71:71:5a:3a:06:44:
                1e:91:e0:80:43:1c:0f:d1:f2:71:26:c2:b5:83:c3:
                c9:cb:ba:86:0e:ba:65:9b:21:0c:8d:3a:30:43:94:
                5e:a9:48:0b:1d:78:50:43:35:f4:d8:28:de:57:de:
                0d:ad:6d:af:f3:a1:a6:a9:a0:fd:73:89:cc:3a:2c:
                f8:ba:f1:69:7c:8d:fc:b2:16:07:78:1c:3b:0b:a2:
                3c:e4:6b:73:a7:a5:19:65:c2:52:8b:c9:19:08:7e:
                5d:f3:db:e0:3c:9d:dd:41:d1:df:25:25:c6:6a:ea:
                72:38:9a:1c:d5:b0:2d:ca:1d:c8:f2:34:ff:1d:b1:
                9f:d8:8b:91:0d:57:84:2a:11:ab:52:90:0f:21:b0:
                27:aa:c9:0e:47:ea:6b:de:c2:17:83:12:b1:3e:e6:
                6f:a6:97:99:2e:04:94:3e:ef:50:cd:44:d5:5d:b9:
                69:1e:f8:e8:0e:de:e5:89:cb:01:a8:be:d0:7b:15:
                ae:ee:44:4c:c3:4c:af:03:98:f1:f0:39:01:04:83:
                bc:73
            Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Key Usage: critical
                Digital Signature, Non Repudiation
            X509v3 Extended Key Usage:
                E-mail Protection
            X509v3 CRL Distribution Points:
                Full Name:
                    URI:http://crl.team14.csa:34535/crl/team14.crl
            X509v3 Subject Key Identifier:
                F1:F8:1E:8E:8D:44:1A:F9:76:21:B7:49:DD:18:2B:6A:64:21:B5:8E
            X509v3 Authority Key Identifier:
                CA:F6:AD:CA:7C:50:C1:61:07:DE:74:42:CF:9A:BF:43:6A:F3:EF:D8
        Signature Algorithm: sha256WithRSAEncryption
        Signature Value:
            68:86:95:15:c2:d8:ab:6c:0b:28:b3:69:99:5d:8b:e6:44:14:
            a4:62:15:57:3a:ef:9c:f0:d6:bd:ce:39:95:72:57:5f:0a:ae:
            09:d5:91:c5:13:1b:6d:81:bd:0c:89:03:8b:72:78:76:20:6c:
            91:69:5d:6f:4a:50:9a:1b:a3:df:4b:6b:95:98:6e:23:36:5a:
            3d:2f:86:d9:f6:c8:c3:cf:51:f9:fa:75:ee:ba:4f:12:37:cb:
            45:5e:ce:af:84:70:96:d2:20:0c:1b:60:cc:91:72:af:0d:0b:
            01:2a:66:b9:fe:5c:1f:fe:58:bb:f2:70:ad:32:47:f0:c8:48:
            b7:dc:1c:05:a7:b3:a4:6c:1a:e7:92:28:cd:b2:ed:ad:16:05:
            67:6b:52:1f:c9:29:ef:9b:fa:f9:9c:25:ab:f6:72:4b:2e:f8:
            f4:5f:fd:f4:aa:22:05:b6:fc:13:41:d3:77:83:d7:ab:76:43:
            23:31:2c:a5:30:60:c5:07:35:00:27:43:a0:b6:c7:7e:a0:49:
            95:42:fd:a7:fd:ad:21:c3:9e:45:7e:bc:de:fe:e8:de:c5:7a:
            f0:79:18:d8:29:8e:66:3e:8f:59:48:bd:6f:cl:eb:22:0d:96:
            5b:f0:2b:7a:f8:76:b6:d8:3e:eb:9d:84:06:33:f6:d7:44:2f:
            67:35:8e:dd:16:0e:d6:4e:d5:a8:89:e9:8d:87:2b:c0:a1:e9:
            05:62:07:e6:1c:e6:7f:0a:53:a3:97:ff:df:de:8d:10:0e:cf:
            74:47:a2:91:65:0a:7e:cb:b8:17:ac:e6:03:f0:0e:f0:7b:bc:
            13:e8:6e:7f:93:3b:95:9d:f1:a1:73:49:ee:1e:12:6d:ed:af:
            7a:fe:47:bb:92:47:1d:6f:59:b5:ac:49:fa:8e:08:8e:18:f1:
            50:cd:2c:e0:24:92:f0:a2:9d:ea:8b:07:d4:2c:1b:52:3d:b7:
            48:34:92:6d:58:59:7d:fc:c7:ab:cb:9a:0d:c7:ee:2c:2c:61:
            7c:2e:7a:43:ae:79:c9:b5:21:1d:09:3c:6f:fd:21:f6:40:63:
            a8:d5:df:22:5b:47:43:6b:b8:37:ae:a2:4c:47:ce:5f:53:6d:
            5d:0d:c6:c3:d0:38:9a:45:c3:22:69:17:55:e2:96:91:ca:21:
            17:03:b2:64:15:39:ce:96:73:d4:4f:ad:0a:79:54:73:ee:fa:
            5c:68:f5:54:9a:40:51:c3:6b:3d:0f:7b:55:c6:e6:d5:61:11:
            2d:5f:a0:35:b8:e9:2d:4d:2b:0c:03:1b:75:3f:bd:98:2a:04:
            7b:59:6e:e5:f3:6c:5d:dc:cc:1e:94:45:ba:e9:b4:17:d3:0d:
            1f:a2:65:06:b1:2d:11:a3
```

Sender encryption certificate details

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Sender-Enc.cer -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 258 (0x102)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = PKI, CN = TEAM14\ROOT, emailAddress = PKI.TEAM14@TEAM14.CA
        Validity
            Not Before: Dec  2 01:21:50 2025 GMT
            Not After : Mar  2 01:21:50 2026 GMT
        Subject: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = TEAM14, CN = TEAM14, emailAddress = SENDER@TEAM14.CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            Public-Key: (2048 bit)
            Modulus:
                00:bf:f2:28:3f:0f:ff:90:bd:11:8c:f6:5a:1c:09:
                d4:72:4c:3a:bb:ef:a2:0a:fc:37:93:a9:80:40:0e:
                f4:cd:3e:22:80:e6:80:7d:b8:ae:af:4d:d1:ce:c9:
                4f:ad:e3:dc:fc:0f:31:c6:56:42:cb:a0:fb:1e:bf:
                59:9c:48:69:5d:13:55:7c:e0:18:2a:f6:58:96:70:
                b0:e0:bb:8b:64:dd:4d:b2:87:21:19:e3:6b:50:7b:
                43:d2:e5:58:aa:f5:64:bd:98:8d:62:fb:00:13:1d:
                4a:2d:6c:2a:18:bb:5f:f6:b2:88:3c:ca:74:21:78:
                53:9d:57:86:f7:19:8e:cd:21:7a:d3:5e:e5:06:ce:
                ed:e2:2d:97:f3:cf:74:f7:db:f5:c4:05:46:3e:14:
                a3:1e:78:b8:48:f9:9d:98:15:c0:5a:7f:5c:e5:d3:
                7e:f3:62:17:a7:9c:9f:e9:b5:e7:eb:24:97:35:14:
                3d:1b:e9:ad:9f:57:07:c5:8b:38:7c:f1:2c:d8:97:
                d5:4f:3b:4f:a4:a1:a7:84:80:88:61:c9:0d:1d:28:
                7a:ae:af:d5:e9:25:01:e4:d1:34:b6:9a:73:b8:86:
                0a:cb:c2:57:3e:3c:47:77:07:d2:a5:0e:0d:10:d9:
                01:15:cf:21:0b:37:c7:67:7d:33:c0:04:15:37:d0:
                96:85
            Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            X509v3 Key Usage: critical
                Key Encipherment
            X509v3 Extended Key Usage:
                E-mail Protection
            X509v3 CRL Distribution Points:
                Full Name:
                    URI:http://crl.team14.csa:34535/crl/crlteam14.crl
            X509v3 Subject Key Identifier:
                2F:6F:A3:5C:5B:30:50:2D:85:9B:69:E1:DA:11:FB:52:6A:13:54:A0
            X509v3 Authority Key Identifier:
                CA:F6:AD:CA:7C:50:C1:61:D7:DE:74:42:CF:9A:BF:43:6A:F3:EF:DB
        Signature Algorithm: sha256WithRSAEncryption
        Signature Value:
            41:42:7a:82:db:7e:07:58:8b:68:81:a7:8c:8b:d8:fb:70:7b:
            c3:b8:6f:b4:b6:fa:95:3d:a2:37:44:e1:da:5c:a1:9a:43:71:
            e7:6b:5a:95:bd:23:de:ee:65:b1:4c:e5:d0:97:c1:e8:b7:b8:
            0c:bd:fe:0f:73:9b:06:d0:79:2b:44:59:92:f4:b9:5b:6a:8a:
            ec:32:5f:05:a0:58:30:6e:93:03:0d:e3:13:39:41:82:ff:1c:
            bc:41:f9:5d:2d:a8:4b:ee:a4:8b:94:3f:91:e9:5b:17:55:1a:
            1e:e0:b3:02:c4:da:07:80:f9:f1:33:37:da:0a:8e:10:a5:1a:
            94:82:95:ea:c3:5c:26:73:d3:38:e6:05:90:33:b7:1a:c7:8c:
            8d:93:31:ad:34:88:7c:06:91:4c:56:89:f7:21:5c:16:18:99:
            7a:6f:1e:00:85:07:29:93:0b:06:42:67:4d:ef:5b:e2:99:65:
            74:26:0e:1a:16:4c:9f:37:e5:42:ce:f4:df:ed:74:08:be:c6:
            eb:f0:e8:c6:d2:fc:89:75:35:73:21:61:7b:69:f0:82:50:5b:
            4c:b7:0e:eb:5d:45:04:2f:1d:34:5e:ae:89:78:c2:e9:6c:7d:
            2a:21:c1:02:eb:18:c7:6e:e4:e0:a4:4d:d0:2d:5a:d1:dd:1a:
            ef:c6:35:a7:59:b0:ce:0a:f0:5b:30:50:bb:28:95:77:a8:7a:
            89:26:33:a3:b4:59:1c:ba:48:04:43:76:dc:dd:bca:d0:09:f4:
            d9:d2:01:50:32:76:62:2a:c7:01:5c:d1:f4:4b:80:ab:d8:d9:
            2d:24:c1:fa:04:85:98:bc:49:aa:65:ab:c3:2a:1e:e6:e5:da:
            79:4a:41:db:e3:61:bf:a4:fa:47:a1:17:8d:87:d8:e9:61:1e:
            b2:fa:ec:fb:c0:0b:c9:2f:b6:e7:6d:16:14:73:ab:cf:aa:c1:
            aa:e3:1d:2f:24:12:ae:62:91:4c:cd:91:ec:49:0d:2c:51:a1:
            66:69:9b:54:3f:a9:d6:cf:49:49:f0:a9:b1:24:0f:4d:9a:63:
            80:8e:82:3f:cc:ee:db:29:dc:7a:ee:65:74:13:a9:4b:e7:dc:
            1a:40:dc:40:9a:9c:23:ac:b8:08:9e:ba:4f:2e:d4:88:42:e3:
            e6:84:32:3c:26:41:7f:58:7b:1e:aa:af:4a:23:ac:73:42:a7:
            6d:db:93:0d:ff:5b:28:42:2c:07:63:3a:51:f8:4e:51:97:34:
            b2:d8:a7:ff:51:bc:93:39:20:8b:73:96:bc:e7:1a:17:29:8a:
            d0:dc:fd:fa:48:6d:49:88:a1:b2:d5:a8:37:d9:97:cf:ff:fb:
            24:7c:93:1c:60:0c:4e:b1
```

Recipient signing certificate details:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Recv-Sign.cer -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number: 513 (0x201)
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = PKI, CN = TEAM14ROOT, emailAddress = PKI.TEAM14@TEAM14.CA
    Validity
      Not Before: Dec  2 01:28:43 2025 GMT
      Not After : Mar  2 01:28:43 2026 GMT
    Subject: C = CA, ST = ONTARIO, L = OTTAWA, O = RECIPIENT, OU = RECIPIENT, CN = RECIPIENT, emailAddress = RECIPIENT@TEAM14.CA
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:9c:db:76:ff:99:94:19:30:7f:ab:be:bc:ae:a4:
        67:08:b9:bb:f7:5d:0d:69:a8:58:5d:c0:e4:a6:fd:
        1b:a0:c0:97:24:8e:66:f7:94:5e:38:93:ab:23:6e:
        8f:ee:a3:97:d4:28:82:88:a6:0b:d5:51:1a:55:8d:
        73:6d:33:b8:fb:48:fd:e2:45:c4:c3:ac:23:82:4c:
        e5:0e:74:45:37:b2:f9:02:a7:a9:61:c2:2b:8f:25:
        ff:7e:e8:a9:d6:7b:46:bf:6d:86:79:d8:30:ad:e2:
        92:fa:83:3e:a1:b7:8c:ed:c5:fb:f7:b7:e3:26:e1:
        80:d2:8a:ee:c8:bb:56:3d:23:fd:59:34:5e:49:32:
        9f:56:2c:2c:4b:ae:4a:4d:90:88:62:de:6a:85:78:
        60:eb:01:bf:89:8a:69:ee:95:0c:29:69:0c:1c:7b:
        cd:50:66:d2:97:e2:6b:93:e2:86:9d:72:cc:b1:b8:
        79:25:4a:cd:5f:3a:eb:02:d1:75:9d:88:d2:ea:a5:
        75:b5:4d:20:99:cd:0a:4f:b5:98:9a:8d:73:f0:8a:
        6d:3a:6c:7e:a0:09:ef:12:43:5d:e9:b8:06:f7:1b:
        76:0f:da:cb:72:ac:72:fd:e0:9b:24:4a:be:52:b5:
        a9:55:5c:9b:47:c2:8f:66:e1:08:b8:9b:95:5f:51:
        1f:19
      Exponent: 65537 (0x10001)
  X509v3 extensions:
    X509v3 Basic Constraints:
      CA:FALSE
    X509v3 Key Usage: critical
      Digital Signature, Non Repudiation
    X509v3 Extended Key Usage:
      E-mail Protection
    X509v3 CRL Distribution Points:
      Full Name:
        URI:http://crl.team14.csa:34535/crl/crlteam14.crl
    X509v3 Subject Key Identifier:
      A4:43:0A:D3:38:04:CB:04:48:C8:88:D7:1F:44:00:B4:AA:19:97:8F
    X509v3 Authority Key Identifier:
      CA:F6:AD:CA:7C:50:C1:61:D7:DE:74:42:CF:9A:BF:43:6A:F3:EF:DB
  Signature Algorithm: sha256WithRSAEncryption
  Signature Value:
    21:73:7f:4d:7a:f6:93:a5:1b:d0:79:5a:07:db:30:00:83:27:
    93:dd:bc:31:e5:9b:dd:2e:12:c7:c1:9e:8c:56:96:df:df:68:
    03:7f:aa:8e:ae:f0:f0:8b:52:dd:64:3f:23:fd:15:55:d9:98:
    40:fd:6c:53:64:87:ea:87:75:f3:64:12:13:9d:a6:b4:b2:e8:
    5e:4a:ac:5c:04:19:49:51:32:29:6d:d1:2e:16:ce:b9:b5:ab:
    3f:3a:7b:53:af:ce:1b:02:93:f5:a0:5e:2e:2d:8d:51:07:b5:
    4a:45:d7:ea:0c:3b:a3:79:b6:20:17:df:20:08:af:27:0d:b8:
    2e:4a:d8:ea:0c:95:d8:f8:7f:b4:93:bd:71:e6:62:46:ad:af:
    db:a4:a9:66:87:ef:c1:df:24:f6:d3:33:b4:1c:34:86:30:1f:
    e5:91:14:ce:2a:70:a5:91:78:0a:44:e1:3b:cb:29:24:85:95:
    c6:08:bc:82:b8:6a:a8:c1:c4:37:2c:81:2a:86:66:7e:c3:fb:
    2e:44:48:7c:c9:5a:68:45:fb:41:cd:c9:3d:33:a1:53:02:55:
    dc:60:23:d1:44:dc:20:7b:d5:7e:e4:9a:58:e6:d3:29:19:60:
    a6:3e:55:47:68:13:65:ab:ce:16:86:47:f9:12:6e:7f:25:c6:
    9e:69:da:cd:4f:56:3b:06:86:9d:45:a4:f1:b6:91:57:a5:3e:
    62:b5:ce:2a:ed:f6:e7:2c:87:c2:38:47:8c:5f:f0:e3:f1:67:
    28:6e:99:96:01:ab:ae:6e:e0:19:b8:3f:69:ac:33:d5:8d:72:
    c1:a5:02:7b:56:76:c4:15:73:94:bf:93:36:df:64:80:34:b3:
    d0:54:66:f5:93:aa:f4:19:3f:b4:2f:d2:5c:d8:d1:0d:da:0c:
    04:de:86:9a:96:d5:62:61:31:fc:cc:71:5b:74:9c:00:fa:c8:
    b8:07:86:e9:7a:5c:ea:1c:ae:a5:06:21:e5:5f:3a:36:20:4b:
    72:54:5c:a2:f4:9d:61:5a:b1:1c:61:bf:6f:eb:4e:41:55:79:
    8c:17:32:51:90:57:84:6b:cc:e8:3a:bd:59:19:27:70:27:7d:
    1e:ec:0b:c0:7a:57:40:4a:9e:16:70:e7:95:b1:81:f8:6f:9c:
    dd:c5:d7:c0:1d:d7:9e:9a:b3:03:cb:bb:c9:5b:f6:b2:b4:43:
    4e:7a:58:3a:33:b2:a7:32:44:68:4e:93:73:c7:6e:0e:2a:dd:
    b9:72:0a:61:b1:e0:da:38:89:ef:17:4d:0c:e4:bc:31:e3:f7:
    82:13:af:90:e5:72:f2:93:5c:82:5c:c9:13:43:eb:ca:70:07:
    12:2c:b0:92:0e:7b:93:9d
```

Recipient encryption certificate details:

```
D:\STUDIES\Crypto\CST8805\PROJECT>openssl x509 -in Recv-Enc.cer -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 514 (0x202)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C = CA, ST = ONTARIO, L = OTTAWA, O = TEAM14, OU = PKI, CN = TEAM14ROOT, emailAddress = PKI.TEAM14@TEAM14.CA
        Validity
            Not Before: Dec  2 01:29:35 2025 GMT
            Not After : Mar  2 01:29:35 2026 GMT
        Subject: C = CA, ST = ONTARIO, L = OTTAWA, O = RECIPIENT, OU = RECIPIENT, CN = ECIPIENT, emailAddress = RECIPIENT@TEAM14.CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            Public-Key: (2048 bit)
            Modulus:
                00:95:ee:8d:5a:3a:fb:a7:e0:61:2d:9c:03:6d:7c:
                e9:3d:cd:09:21:85:78:ee:44:fe:e3:40:9c:94:62:
                52:12:6e:f9:8e:af:c4:5c:f4:a7:a1:94:78:93:d9:
                ee:da:8a:1a:5b:22:dc:23:7c:9a:98:c6:96:7e:86:
                e5:6d:21:ab:5d:11:7c:b0:e1:56:b4:37:23:c4:87:
                78:a5:76:2a:6a:0a:3c:78:3b:f3:23:49:94:07:1b:
                06:9b:4a:64:47:10:1d:4e:e9:02:59:b5:69:17:5d:
                56:11:4f:2f:e0:59:8a:0c:f8:33:19:04:45:88:f5:
                15:09:0d:3c:ed:86:bf:5e:cf:ae:47:a9:35:30:c1:
                4f:af:3b:68:e4:b1:8e:92:2a:2f:36:91:58:19:a3:
                ae:72:b5:a2:23:5f:c8:b7:91:06:7f:7a:30:03:57:
                e1:39:31:ee:6f:30:90:c9:7a:ba:32:6b:a3:f0:3c:
                22:de:15:57:21:28:41:a2:15:d9:d6:0f:77:a4:6a:
                38:04:18:a2:84:01:29:52:fa:08:9f:ec:4a:74:5f:
                15:21:d7:01:ae:8b:a8:56:33:0d:f3:cd:4f:70:57:
                51:b8:fc:a7:7c:bd:b5:41:9f:78:06:a2:3d:05:af:
                ce:cf:2d:c9:00:a4:b0:fc:39:c7:c5:cd:84:b4:3f:
                db:71
            Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            X509v3 Key Usage: critical
                Key Encipherment
            X509v3 Extended Key Usage:
                E-mail Protection
            X509v3 CRL Distribution Points:
                Full Name:
                    URI:http://crl.team14.csa:34535/crl/crlteam14.crl
            X509v3 Subject Key Identifier:
                38:E8:F1:B0:30:11:46:93:B3:B0:A8:19:8B:3C:11:0E:6E:EF:4E:29
            X509v3 Authority Key Identifier:
                CA:F6:AD:CA:7C:50:C1:61:D7:DE:74:42:CF:9A:BF:43:6A:F3:EF:DB
        Signature Algorithm: sha256WithRSAEncryption
        Signature Value:
            56:69:21:be:13:94:2b:7b:a3:73:96:9e:ec:dd:d2:75:3d:dc:
            e6:4b:3b:c4:4a:53:58:3b:62:79:cc:4a:2c:4e:e4:92:01:4e:
            47:74:4d:d7:11:8e:c6:eb:70:f8:30:89:04:b8:ed:1a:6c:b3:
            c8:72:dd:0a:c2:27:5b:c2:7e:f0:68:4b:3c:78:65:0b:d7:57:
            45:e9:e9:73:4b:71:83:f0:78:a7:e2:70:82:88:e2:be:6a:
            d5:aa:c1:d3:4d:e5:7e:38:ed:6c:55:39:87:06:9f:e5:16:9c:
            11:d6:01:66:59:ce:65:4c:1e:ab:a4:29:3e:c0:7d:bc:63:c2:
            9c:94:a9:3e:84:33:ad:df:e2:6d:a3:38:cd:a1:05:ae:03:b5:
            c6:6c:a3:3c:ae:81:40:ff:da:a5:4d:6c:8f:42:77:54:1a:76:
            56:d6:7d:ec:8a:08:a3:01:e0:a0:6d:23:66:fd:d3:aa:42:69:
            2e:9c:f5:56:47:1a:d2:73:41:0c:77:02:99:e9:42:e2:70:7a:
            af:0b:04:50:33:4b:37:c0:2f:cb:e0:b3:d6:e3:89:23:f3:68:
            83:05:0f:a4:6b:b8:c6:be:38:61:ed:83:cd:59:ec:5b:81:52:
            4e:05:52:3d:a9:7d:54:62:77:5d:f7:4e:22:a0:dc:6d:92:96:
            a9:a6:32:44:e6:25:09:ce:0d:0a:90:a9:f8:82:4d:2b:98:6f:
            74:d8:71:a3:d1:ce:fe:a3:0f:2a:98:9a:5f:eb:46:bf:e3:e4:
            96:0b:c8:30:36:d4:4c:fb:0b:30:d0:27:fc:67:96:bf:05:f5:
            56:54:2c:1e:66:a6:f3:28:5a:59:b9:6f:66:ba:95:d3:3d:1b:
            d0:b3:3b:5b:0f:a5:f5:ce:68:7c:72:20:2d:52:93:c2:69:1f:
            c8:fb:f3:bd:0e:fb:34:d6:1f:75:4c:e0:57:a5:96:52:13:c0:
            62:1f:0d:41:46:78:77:50:40:04:40:52:c4:c0:d3:0c:67:c0:
            a4:3e:c6:42:fb:bb:ae:86:5b:1d:d8:4f:5c:7d:e6:fb:f5:da:
            27:6c:bc:5f:34:10:80:6e:84:3e:e3:38:1d:3f:9d:6a:0f:c9:
            70:1f:be:07:5f:f6:0a:d3:25:51:4f:56:97:1d:53:93:0c:5c:
            6a:af:45:9f:05:ba:b1:df:84:16:dd:ec:f1:1a:94:cb:80:75:
            af:69:65:9b:fa:b1:4f:a6:cf:55:00:f0:a9:45:a1:e7:16:75:
            bb:34:fc:ba:a9:4e:54:b2:e0:8e:3c:e9:13:1d:ed:c5:84:00:
            f3:06:0a:20:aa:8c:24:41:9a:ac:d8:a3:13:a6:22:9a:e0:21:
            ee:25:f2:63:65:06:7f:07
```

Certificate Chain & Trusted Root

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl verify -verbose -CAfile CArootCert.cer Sender-Sign.cer
Sender-Sign.cer: OK

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl verify -verbose -CAfile CArootCert.cer Sender-Enc.cer
Sender-Enc.cer: OK

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl verify -verbose -CAfile CArootCert.cer Recv-Sign.cer
Recv-Sign.cer: OK

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl verify -verbose -CAfile CArootCert.cer Recv-Enc.cer
Recv-Enc.cer: OK
```

CRL Revoke:

```
C:\STUDIES\Crypto\CS7889\PROJECT2\openssl verify -verbose c:\_chx\chx-file C:\Project\rsl_pos -Cafile C:\rootCert.cac untrusted.cac  
C = CA, ST = ON, L = Ottawa, O = UNTRUSTED, OU = UNTRUSTED, emailAddress = untrusted@untrusted.ca  
error 16 at depth lookup: unable to get certificate 0:  
C = CA, ST = ON, L = Ottawa, O = UNTRUSTED, OU = UNTRUSTED, emailAddress = untrusted@untrusted.ca  
error 3 at depth lookup: unable to get certificate CR.  
error untrusted.cac verification failed  
40800000:error:16000000:STORE routines:ssl_store_get_loader_inner:unregistered scheme:cryptostore\store_register.c:237:schemefile  
cryptostore\library_files\bin\msi\capi_x64 directory does not exist: cryptostore\library_files\bin\msi\capi_x64 directory does not exist: calling stat(C:\Program Files\Common Files\SSL\certs)  
40800000:error:16000000:STORE routines:ssl_store_get_loader_inner:unregistered scheme:cryptostore\store_register.c:237:schemefile  
40800000:error:16000000:STORE routines:inner_loader_fetch_unsupported:cryptostore\store_meth.c:383>No store loader found. For standard store loaders you need at least one of the default or base providers available. Did you forget to load them? Info: Global default library context, Scheme (C : 0), Properties (None)  
40800000:error:16000000:STORE routines:ssl_store_get_loader_inner:unregistered scheme:cryptostore\store_register.c:237:schemefile  
cryptostore\library_files\bin\msi\capi_x64 directory does not exist: cryptostore\library_files\bin\msi\capi_x64 directory does not exist: calling stat(C:\Program Files\Common Files\SSL\certs)  
40800000:error:16000000:STORE routines:ssl_store_get_loader_inner:unregistered scheme:cryptostore\store_register.c:237:schemefile  
40800000:error:16000000:STORE routines:inner_loader_fetch_unsupported:cryptostore\store_meth.c:383>No store loader found. For standard store loaders you need at least one of the default or base providers available. Did you forget to load them? Info: Global default library context, Scheme (C : 0), Properties (None)  
(1)
```

Negative Tests:

Cert Verification Failed:

[illegible]

NSIT TESTS:

```
D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in CARootCert.cer -text -noout | findstr /i "Public-Key Signature Algorithm"
    Signature Algorithm: sha256WithRSAEncryption
      Public Key Algorithm: rsaEncryption
        Public-Key: (4096 bit)
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Sender-Sign.cer -text -noout | findstr /i "Public-Key Signature Algorithm"
    Signature Algorithm: sha256WithRSAEncryption
      Public Key Algorithm: rsaEncryption
        Public-Key: (2048 bit)
      Digital Signature, Non Repudiation
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Sender-Enc.cer -text -noout | findstr /i "Public-Key Signature Algorithm"
    Signature Algorithm: sha256WithRSAEncryption
      Public Key Algorithm: rsaEncryption
        Public-Key: (2048 bit)
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Recv-Sign.cer -text -noout | findstr /i "Public-Key Signature Algorithm"
    Signature Algorithm: sha256WithRSAEncryption
      Public Key Algorithm: rsaEncryption
        Public-Key: (2048 bit)
      Digital Signature, Non Repudiation
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:

D:\STUDIES\Crypto\CST8805\PROJECT2>openssl x509 -in Recv-Enc.cer -text -noout | findstr /i "Public-Key Signature Algorithm"
    Signature Algorithm: sha256WithRSAEncryption
      Public Key Algorithm: rsaEncryption
        Public-Key: (2048 bit)
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
```